



BA, BSc Degree Examinations

Department:

School for Business and Society

Title of Exam:

50% Closed Exam

Time Allowed:

1 hour 30 minutes

Allocation of Marks:

Marks are allocated for each question in the main body of the exam.

Instructions for Candidates:

Answer all questions.

Where applicable provide a narrative, formula and show your workings. Answer to two decimal places.

CONTINUE TO NEXT PAGE

Question 1:

Table 1 shows the number of hours that a group of university students spent studying for their final exams. Each student followed their own schedule and study plan.

Table 1

	Study Hours
Emma	45
James	38
Olivia	52
Daniel	44
Sophia	39
Ethan	49
Mia	41
Lucas	50
Ava	42
Noah	55
Harper	37
Caleb	40
Scarlet	46
Owen	48
Zoe	43

Required:

- a) Provide a **five-number summary** of the study hours of the students presented in Table 1.
(10 marks)
- b) Draw by hand, a box and whisker plot for the data using relevant statistics calculated in part a).
(5 marks)
- c) Calculate the interquartile range of the data.
(5 marks)
- d) Calculate the standard deviation of the study hours of the students.
(10 marks)

TOTAL: 30 marks

Question 2:

A bakery produces two types of pastries—Croissants and Danish Pastries. Information on the number of regular and chocolate-filled pastries for each type is provided in Table 2. Imagine you randomly select a single pastry from a tray.

Table 2

Type of Pastry	Regular	Chocolate-Filled
Croissant	80	150
Danish	50	170

Required:

- a) Calculate the probability that a randomly selected pastry is a **Croissant**.
(5 marks)
- b) Calculate the probability that a randomly selected pastry is **Regular**.
(5 marks)
- c) Calculate the probability that a randomly selected pastry is a **Croissant**, given that it is **Chocolate-Filled**.
(5 marks)
- d) Calculate the probability that a randomly selected pastry is either **Chocolate-Filled** or a **Danish Pastry**.

(5 marks)

TOTAL: 20 marks

Question 3:

A warehouse security system has been installed to detect theft attempts. The system relies on motion sensors and AI-based monitoring to trigger an alert when suspicious activity is detected. However, the system is not perfect.

Based on historical data:

- The probability that a genuine theft attempt occurs is **0.07**.
- If a theft is actually happening, the probability that the security system triggers an alert (true positive) is **0.92**.
- If no theft is occurring, the probability that the system still triggers an alert (false positive) is **0.15**.

What is the probability that **no theft is occurring, given that the security system has triggered an alert?**

TOTAL: 15 marks

Question 4:

A company is considering launching a new product, and an analyst estimates the returns in three different market scenarios. The projected returns are £200, £80, and £100. The analyst estimates the probabilities of the first two scenarios as 0.20 and 0.35, respectively, with the third scenario happening the rest of the time. What is the expected return from the product launch?

TOTAL: 5 marks

Question 5:

In a certain city, **25%** of households own an electric car. Assuming that each household is independent and identically distributed:

- a) Calculate the probability that in a group of **12** households, exactly **3** own an electric car.
(5 marks)
- b) Calculate the probability that in a group of **12** households, there are at least **3** households that own an electric car.
(10 marks)
- c) If we model the probability of a household owning an electric car using a Bernoulli distribution, calculate the **mean** and **variance**.
(5 marks)

TOTAL: 20 marks

Question 6:

A bakery evaluates the weight of loaves of bread using a sample of **10** loaves. The weight of each loaf (in kilograms) is provided in Table 3.

Table 3

Loaf	Weight (kg)
1	0.52
2	0.50
3	0.54
4	0.55
5	0.51
6	0.53
7	0.56
8	0.50
9	0.52
10	0.55

Required:

- a) Calculate the **mean** and **standard deviation** for the sample. (5 marks)
- b) Determine the 98% confidence interval for the true mean of the population weight. (5 marks)

TOTAL: 10 marks**END OF EXAMINATION**

Some Formulae and Tables:

Parameter	Bernoulli	Binomial
Probability	$p^r (1 - p)^{(1-r)}$	${}^nC_r p^r (1 - p)^{(n-r)}$
Mean	p	np
Variance	$p(1 - p)$	$np(1 - p)$

Sample Sizes:

$$n = \left(\frac{CC^2 p(1-p)}{a^2} \right) \text{ or } n = \left(\frac{CCs}{a} \right)^2$$

Standard Error of the Mean:

$$SEM = \frac{s}{\sqrt{n}}$$

Testing two means with known σ or a large sample:

$$\text{Test Statistic} = \frac{|\bar{x}_1 - \bar{x}_2|}{STED} \text{ where } STED = \sqrt{\left(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} \right)} \text{ with } v = n_1 + n_2 - 2$$

Testing two means with unknown σ and a small sample:

$$\text{Test Statistic} = \frac{|\bar{x}_1 - \bar{x}_2|}{STED} \text{ where } STED = s_p \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

$$\text{and } s_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}} \text{ with } v = n_1 + n_2 - 2$$

Testing two proportions

$$\text{Test Statistic} = \frac{|p_1 - p_2|}{STED} \text{ where } STED = \sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

$$\text{and } \hat{p} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} \text{ with } v = n_1 + n_2 - 2$$

Chi-Squared Test:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

PERCENTAGE POINTS OF THE T-DISTRIBUTION

One Tail	5%	2.5%	1%	0.5%	0.1%	0.05%
Two Tails	10%	5%	2%	1%	0.2%	0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

Standard Normal Distribution Tables: Values represent AREA to the LEFT of z.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.90	0.00005	0.00005	0.00004	0.00004	0.00004	0.00004	0.00004	0.00004	0.00003	0.00003
-3.80	0.00007	0.00007	0.00007	0.00006	0.00006	0.00006	0.00006	0.00005	0.00005	0.00005
-3.70	0.00011	0.00010	0.00010	0.00010	0.00009	0.00009	0.00008	0.00008	0.00008	0.00008
-3.60	0.00016	0.00015	0.00015	0.00014	0.00014	0.00013	0.00013	0.00012	0.00012	0.00011
-3.50	0.00023	0.00022	0.00022	0.00021	0.00020	0.00019	0.00019	0.00018	0.00017	0.00017
-3.40	0.00034	0.00032	0.00031	0.00030	0.00029	0.00028	0.00027	0.00026	0.00025	0.00024
-3.30	0.00048	0.00047	0.00045	0.00043	0.00042	0.00040	0.00039	0.00038	0.00036	0.00035
-3.20	0.00069	0.00066	0.00064	0.00062	0.00060	0.00058	0.00056	0.00054	0.00052	0.00050
-3.10	0.00097	0.00094	0.00090	0.00087	0.00084	0.00082	0.00079	0.00076	0.00074	0.00071
-3.00	0.00135	0.00131	0.00126	0.00122	0.00118	0.00114	0.00111	0.00107	0.00104	0.00100
-2.90	0.00187	0.00181	0.00175	0.00169	0.00164	0.00159	0.00154	0.00149	0.00144	0.00139
-2.80	0.00256	0.00248	0.00240	0.00233	0.00226	0.00219	0.00212	0.00205	0.00199	0.00193
-2.70	0.00347	0.00336	0.00326	0.00317	0.00307	0.00298	0.00289	0.00280	0.00272	0.00264
-2.60	0.00466	0.00453	0.00440	0.00427	0.00415	0.00402	0.00391	0.00379	0.00368	0.00357
-2.50	0.00621	0.00604	0.00587	0.00570	0.00554	0.00539	0.00523	0.00508	0.00494	0.00480
-2.40	0.00820	0.00798	0.00776	0.00755	0.00734	0.00714	0.00695	0.00676	0.00657	0.00639
-2.30	0.01072	0.01044	0.01017	0.00990	0.00964	0.00939	0.00914	0.00889	0.00866	0.00842
-2.20	0.01390	0.01355	0.01321	0.01287	0.01255	0.01222	0.01191	0.01160	0.01130	0.01101
-2.10	0.01786	0.01743	0.01700	0.01659	0.01618	0.01578	0.01539	0.01500	0.01463	0.01426
-2.00	0.02275	0.02222	0.02169	0.02118	0.02068	0.02018	0.01970	0.01923	0.01876	0.01831
-1.90	0.02872	0.02807	0.02743	0.02680	0.02619	0.02559	0.02500	0.02442	0.02385	0.02330
-1.80	0.03593	0.03515	0.03438	0.03362	0.03288	0.03216	0.03144	0.03074	0.03005	0.02938
-1.70	0.04457	0.04363	0.04272	0.04182	0.04093	0.04006	0.03920	0.03836	0.03754	0.03673
-1.60	0.05480	0.05370	0.05262	0.05155	0.05050	0.04947	0.04846	0.04746	0.04648	0.04551
-1.50	0.06681	0.06552	0.06426	0.06301	0.06178	0.06057	0.05938	0.05821	0.05705	0.05592
-1.40	0.08076	0.07927	0.07780	0.07636	0.07493	0.07353	0.07215	0.07078	0.06944	0.06811
-1.30	0.09680	0.09510	0.09342	0.09176	0.09012	0.08851	0.08691	0.08534	0.08379	0.08226
-1.20	0.11507	0.11314	0.11123	0.10935	0.10749	0.10565	0.10383	0.10204	0.10027	0.09853
-1.10	0.13567	0.13350	0.13136	0.12924	0.12714	0.12507	0.12302	0.12100	0.11900	0.11702
-1.00	0.15866	0.15625	0.15386	0.15151	0.14917	0.14686	0.14457	0.14231	0.14007	0.13786
-0.90	0.18406	0.18141	0.17879	0.17619	0.17361	0.17106	0.16853	0.16602	0.16354	0.16109
-0.80	0.21186	0.20897	0.20611	0.20327	0.20045	0.19766	0.19489	0.19215	0.18943	0.18673
-0.70	0.24196	0.23885	0.23576	0.23270	0.22965	0.22663	0.22363	0.22065	0.21770	0.21476
-0.60	0.27425	0.27093	0.26763	0.26435	0.26109	0.25785	0.25463	0.25143	0.24825	0.24510
-0.50	0.30854	0.30503	0.30153	0.29806	0.29460	0.29116	0.28774	0.28434	0.28096	0.27760
-0.40	0.34458	0.34090	0.33724	0.33360	0.32997	0.32636	0.32276	0.31918	0.31561	0.31207
-0.30	0.38209	0.37828	0.37448	0.37070	0.36693	0.36317	0.35942	0.35569	0.35197	0.34827
-0.20	0.42074	0.41683	0.41294	0.40905	0.40517	0.40129	0.39743	0.39358	0.38974	0.38591
-0.10	0.46017	0.45620	0.45224	0.44828	0.44433	0.44038	0.43644	0.43251	0.42858	0.42465
0.00	0.50000	0.49601	0.49202	0.48803	0.48405	0.48006	0.47608	0.47210	0.46812	0.46414

Standard Normal Distribution Tables: Values represent AREA to the LEFT of z.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.00	0.50000	0.50399	0.50798	0.51197	0.51595	0.51994	0.52392	0.52790	0.53188	0.53586
0.10	0.53983	0.54380	0.54776	0.55172	0.55567	0.55962	0.56356	0.56749	0.57142	0.57535
0.20	0.57926	0.58317	0.58706	0.59095	0.59483	0.59871	0.60257	0.60642	0.61026	0.61409
0.30	0.61791	0.62172	0.62552	0.62930	0.63307	0.63683	0.64058	0.64431	0.64803	0.65173
0.40	0.65542	0.65910	0.66276	0.66640	0.67003	0.67364	0.67724	0.68082	0.68439	0.68793
0.50	0.69146	0.69497	0.69847	0.70194	0.70540	0.70884	0.71226	0.71566	0.71904	0.72240
0.60	0.72575	0.72907	0.73237	0.73565	0.73891	0.74215	0.74537	0.74857	0.75175	0.75490
0.70	0.75804	0.76115	0.76424	0.76730	0.77035	0.77337	0.77637	0.77935	0.78230	0.78524
0.80	0.78814	0.79103	0.79389	0.79673	0.79955	0.80234	0.80511	0.80785	0.81057	0.81327
0.90	0.81594	0.81859	0.82121	0.82381	0.82639	0.82894	0.83147	0.83398	0.83646	0.83891
1.00	0.84134	0.84375	0.84614	0.84849	0.85083	0.85314	0.85543	0.85769	0.85993	0.86214
1.10	0.86433	0.86650	0.86864	0.87076	0.87286	0.87493	0.87698	0.87900	0.88100	0.88298
1.20	0.88493	0.88686	0.88877	0.89065	0.89251	0.89435	0.89617	0.89796	0.89973	0.90147
1.30	0.90320	0.90490	0.90658	0.90824	0.90988	0.91149	0.91309	0.91466	0.91621	0.91774
1.40	0.91924	0.92073	0.92220	0.92364	0.92507	0.92647	0.92785	0.92922	0.93056	0.93189
1.50	0.93319	0.93448	0.93574	0.93699	0.93822	0.93943	0.94062	0.94179	0.94295	0.94408
1.60	0.94520	0.94630	0.94738	0.94845	0.94950	0.95053	0.95154	0.95254	0.95352	0.95449
1.70	0.95543	0.95637	0.95728	0.95818	0.95907	0.95994	0.96080	0.96164	0.96246	0.96327
1.80	0.96407	0.96485	0.96562	0.96638	0.96712	0.96784	0.96856	0.96926	0.96995	0.97062
1.90	0.97128	0.97193	0.97257	0.97320	0.97381	0.97441	0.97500	0.97558	0.97615	0.97670
2.00	0.97725	0.97778	0.97831	0.97882	0.97932	0.97982	0.98030	0.98077	0.98124	0.98169
2.10	0.98214	0.98257	0.98300	0.98341	0.98382	0.98422	0.98461	0.98500	0.98537	0.98574
2.20	0.98610	0.98645	0.98679	0.98713	0.98745	0.98778	0.98809	0.98840	0.98870	0.98899
2.30	0.98928	0.98956	0.98983	0.99010	0.99036	0.99061	0.99086	0.99111	0.99134	0.99158
2.40	0.99180	0.99202	0.99224	0.99245	0.99266	0.99286	0.99305	0.99324	0.99343	0.99361
2.50	0.99379	0.99396	0.99413	0.99430	0.99446	0.99461	0.99477	0.99492	0.99506	0.99520
2.60	0.99534	0.99547	0.99560	0.99573	0.99585	0.99598	0.99609	0.99621	0.99632	0.99643
2.70	0.99653	0.99664	0.99674	0.99683	0.99693	0.99702	0.99711	0.99720	0.99728	0.99736
2.80	0.99744	0.99752	0.99760	0.99767	0.99774	0.99781	0.99788	0.99795	0.99801	0.99807
2.90	0.99813	0.99819	0.99825	0.99831	0.99836	0.99841	0.99846	0.99851	0.99856	0.99861
3.00	0.99865	0.99869	0.99874	0.99878	0.99882	0.99886	0.99889	0.99893	0.99896	0.99900
3.10	0.99903	0.99906	0.99910	0.99913	0.99916	0.99918	0.99921	0.99924	0.99926	0.99929
3.20	0.99931	0.99934	0.99936	0.99938	0.99940	0.99942	0.99944	0.99946	0.99948	0.99950
3.30	0.99952	0.99953	0.99955	0.99957	0.99958	0.99960	0.99961	0.99962	0.99964	0.99965
3.40	0.99966	0.99968	0.99969	0.99970	0.99971	0.99972	0.99973	0.99974	0.99975	0.99976
3.50	0.99977	0.99978	0.99978	0.99979	0.99980	0.99981	0.99981	0.99982	0.99983	0.99983
3.60	0.99984	0.99985	0.99985	0.99986	0.99986	0.99987	0.99987	0.99988	0.99988	0.99989
3.70	0.99989	0.99990	0.99990	0.99990	0.99991	0.99991	0.99992	0.99992	0.99992	0.99992
3.80	0.99993	0.99993	0.99993	0.99994	0.99994	0.99994	0.99994	0.99995	0.99995	0.99995
3.90	0.99995	0.99995	0.99996	0.99996	0.99996	0.99996	0.99996	0.99996	0.99997	0.99997